MATH 112B

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Spring 2026

Contents

1 Diffusion & Wave

1.1 Lecture 1 - Gradient Systems

<u>Notation</u>: for a function with two variables

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$
$$\frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

Linearity.

$$\frac{\partial}{\partial x} cf = c\frac{\partial}{\partial x} f \tag{1}$$
$$\frac{\partial}{\partial x}(f+g) = \frac{\partial}{\partial x} f + \frac{\partial}{\partial x} g \tag{2}$$

Gradient. For $f = f(x,y)$, define the gradient as the following:

$$\vec{\nabla} f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

<u>Properties</u>:

1. Magnitude: $|\vec{\nabla} f| = \sqrt{f_x^2 + f_y^2}$
2. Direction: $\vec{u} = \frac{\vec{\nabla} f}{|\vec{\nabla} f|}$ = unit vector in the $(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ direction.
3. $\vec{\nabla} f$ at point (x_0, y_0) is $\perp$ to the level set at point (x_0, y_0).
4. $\vec{\nabla} f$ is defined at all the points (x, y) where partials exists.
5. A collection of all $\vec{\nabla} f$ is a vector field called gradient field.
6. One could verify $F(M(x,y), N(x,y))$ is a gradient field by verifying

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

   where $F = \vec{\nabla} f$

1.2 Lecture 2 - Conservation Laws

Conservation Laws

This is used to describe: bacterial movement in a petridish, nerve signal traveling through an axon, heat diffusion in a rod, cars on the highway.

We start with the following assumptions:

- Motion happens in 1D.
- Cross-sectional area is constant.

Consider a 1D tube, with particles inside. How do we describe the change of concentration of particles?

$$\begin{array}{c}\text{change in pop between}\\ x \text{ and } x+\Delta x\end{array} = \begin{array}{c}\text{rate of entry into}\\ (x, x+\Delta x)\\ \text{per unit time}\end{array} - \begin{array}{c}\text{rate of exit into}\\ (x, x+\Delta x)\\ \text{per unit time}\end{array}$$

Define

- $c(x,t)$ = concentration of particles at x at time t
- $J(x,t)$ = flux of particles at (x,t) = # of particles crossing a unit area at x in the positive direction, per unit time.
- A = cross-sectional area of the tube.
- ΔV = volume of our segment = $A \cdot \Delta x$

We have

$$\begin{aligned}\frac{\partial}{\partial t}(\underbrace{c(x,t)\cdot A\cdot \Delta x}_{\#\text{ of particles}}) &= J(x,t)A - J(x+\Delta x,t)A\\ \frac{\partial c}{\partial t} &= \frac{J(x,t)-J(x+\Delta x,t)}{\Delta x}\\ \lim_{\Delta x\to 0}\frac{\partial c}{\partial t} &= \lim_{\Delta x\to 0}\frac{J(x,t)-J(x+\Delta x,t)}{\Delta x}\\ \frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial x} \leftarrow \text{Conservation of mass}\end{aligned}$$

Consider a new term,

$$\begin{array}{c} \Delta\text{pop between} \\ x \text{ and } x+\Delta x \end{array} = \begin{array}{c} \text{rate of entry into} \\ (x, x+\Delta x) \\ \text{per unit time} \end{array} - \begin{array}{c} \text{rate of exit into} \\ (x, x+\Delta x) \\ \text{per unit time} \end{array} \pm \begin{array}{c} \text{rate of local} \\ \text{creation} \\ \text{or degradation} \end{array}$$

We now have

- $\sigma(x,t) = \#$ of particles created/removed per unit volume at time t

Now our conservation laws become

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} \pm \sigma(x,t)$$

To generalize to 2D or 3D:

$$\begin{aligned} \frac{\partial c}{\partial t} &= -\vec{\nabla}\vec{J} \pm \sigma(\vec{x},t) \\ \text{Now we have } c &= c(\vec{x},t) \\ \sigma &= \sigma(\vec{x},t) \\ \vec{J} &= (J_x, J_y, J_z) \\ \vec{\nabla} &= (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}) \end{aligned}$$

So we have

$$\frac{\partial c}{\partial t} = -(\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}) \pm \sigma(x,y,z,t)$$

$\boxed{\text{ex}}$ Consider a moving fluid + organisms that move with the flow (not actively). Suppose that the velocity of the fluid is given by

$$\vec{v}(x,y,z,t)$$

Let $c(x,y,z,t) =$ concentration of organism.

What is the flux?

$$\begin{aligned} \vec{J} &= c \cdot \vec{V} \quad \text{Assume } \sigma = 0 \\ \Rightarrow \frac{\partial c}{\partial t} &= -\left(\frac{\partial cV_x}{\partial x} + \frac{\partial cV_y}{\partial y} + \frac{\partial cV_z}{\partial z}\right) \end{aligned}$$

In 1D:

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}(cV) \\ &= -\frac{\partial c}{\partial x}V - \frac{\partial V}{\partial x}c\end{aligned}$$

What kind of PDE is this?

- 1st order
- 2 variables: x, t
- Linear
- Transport equation

$\boxed{\text{ex}}$ Attraction/repulsion. Suppose that ψ is a source of attraction for the bacteria. The motion is determined by the gradient of ψ.
In 1D: $\psi = \psi(x,t),\ J = \frac{\partial \psi}{\partial x}$

$$J = c\alpha\vec{\nabla}\psi$$

We have

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial X} \\ \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial t}\left(c\alpha\frac{\partial \psi}{\psi x}\right)\end{aligned}$$

ψ is given.

$$\frac{\partial c}{\partial t} = -\alpha\frac{\partial c}{\partial x}\cdot\frac{\partial \psi}{\partial x} - \alpha c\frac{\partial c}{\partial x}\cdot\frac{\partial^2 \psi}{\partial x^2}$$

This is 1st order, linear PDE in 2 variables.

$\boxed{\text{ex}}$ Temperature of a heated rod.
Here we define $c(x,t)$ as concentration of energy/temperature. And since heat spreads from hot to cold, we get the following.

$$\boxed{\text{Fick's Law} : J = -D\cdot\frac{\partial c}{\partial x}}$$

Then we have

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}\left(-D\frac{\partial c}{\partial x}\right)\\ &= D\frac{\partial^2 c}{\partial x^2} \ \leftarrow \text{Heat equation}\end{aligned}$$

In higher dimensions:

$$\begin{aligned}\frac{\partial c}{\partial t} &= \vec{\nabla}(D\vec{\nabla}c)\\ &= D\left(\frac{\partial^2 c}{\partial x^2}+\frac{\partial^2 c}{\partial y^2}+\frac{\partial^2 c}{\partial z^2}\right)\end{aligned}$$

$\boxed{\text{ex}}$ Random motion of particles.
Denote λ_r as the amount of particles jumping from $x \to x+\Delta x$
Denote λ_l as the amount of particles jumping from $x \to x-\Delta x$
Denote $c(x,t) = \#$ of particles within $(x, x+\Delta x)$ at time t:

$$c(x,t+\Delta t) = c(x,t) + \lambda_r \cdot c(x-\Delta x,t) + \lambda_l \cdot c(x+\Delta x,t) - \lambda_l c(x,t) - \lambda_r c(x,t)$$

Using Taylor Series:

$$\begin{aligned}c(x,t+\Delta t) &= c(x,t) + \frac{\partial c}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 c}{\partial t^2}(\Delta t)^2 + \dots\\ c(x+\Delta x,t) &= c(x,t) + \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2 + \dots\\ c(x-\Delta x,t) &= c(x,t) - \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2 + \dots\end{aligned}$$

We can simplify using $\lambda_r = \lambda_l = \lambda$, then we have

$$\begin{aligned}c + \frac{\partial c}{\partial t}\Delta t + \frac{\partial^2 c}{\partial t^2}\frac{(\Delta t)^2}{2} + \dots = c\\ &+ \lambda\left(c + \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2\right)\\ &+ \lambda\left(c - \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2\right)\\ &- \lambda c - \lambda c\\ \frac{\partial c}{\partial t}\Delta t + \frac{\partial^2 c}{\partial t^2}\frac{(\Delta t)^2}{2} &= \lambda\frac{\partial^2 c}{\partial x^2}(\Delta x)^2\end{aligned}$$

Call $\lim_{\Delta x \to 0, \Delta t \to 0} \frac{(\Delta x)^2}{\Delta t} \equiv K$ and ignore second order time term as we take the limit.

$$\frac{\partial c}{\partial t}\Delta t = \lambda(\Delta x)^2 \cdot \frac{\partial^2 c}{\partial x^2}$$
$$\frac{\partial c}{\partial t} = \lambda K \cdot \frac{\partial^2 c}{\partial x^2} \leftarrow \text{Diffusion equation}$$

1.3 Lecture 3 - Conservation Laws cont.

Recall

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} + \sigma(x,t)$$

$\boxed{\text{ex}}$ Propagation of Action Potential along an axon (neural cell). Let

- $x =$ the distance along the axon.
- $q(x,t) =$ the charge density per unit length.
- $J(x,t) =$ flux of charged particles
- $\sigma(x,t) =$ rate of charge entering/leaving

We have the following transport equation for the charge:

$$\frac{\partial q}{\partial t} = -\frac{\partial J}{\partial x} + \sigma(x,t)$$

So what is $J(x,t)$? It is the electrical current.

For $J(x,t)$, we use Ohm's Law:

$$J = -a\frac{\partial V}{\partial x}$$

where a is a constant and $\frac{\partial V}{\partial x}$ is the voltage gradient.

This relates charge to voltage:

$$q - bV$$

where q is charge, b is a constant, V is voltage.

We then have

$$\frac{\partial}{\partial t}(bV) = -\frac{\partial}{\partial x}(-a\frac{\partial V}{\partial x}) + \sigma(x,t)\cdot\frac{1}{b}$$
$$\underbrace{\frac{\partial V}{\partial t} = \frac{a}{b}\frac{\partial^2 V}{\partial x^2}}_{\text{diffusion eq}} + \frac{\sigma(x,t)}{b}$$

Generalize this to 2d or 3d

$$\frac{\partial c}{\partial t} = D\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2}\right) \qquad \text{(diffusion eq)}$$

<u>Del operator</u>: (note that $\vec{\nabla}u = \text{grad } u$)

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$$
$$u(x,y,z) \to \vec{\nabla} \to \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)$$
$$\text{scalar} \to \vec{\nabla} \to \text{vector}$$

<u>Divergence</u>:
Consider the dot product with $\vec{V} = (V_x, V_y, V_z)$, we have

$$\vec{\nabla}\cdot\vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = \text{divergence of } V$$
$$\text{vector} \to \text{div} \to \text{scalar}$$

<u>Laplacian</u>:
Consider the combination of div and grad

$$\vec{\nabla}(\vec{\nabla}u)$$

where $\vec{\nabla}u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)$. We have

$$\underbrace{\vec{\nabla}\cdot\left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)}_{\text{div}\,\cdot\,\text{grad } u} = \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = \underbrace{\Delta u}_{\text{Laplacian } u}$$
$$\Delta = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)$$

Back to the diffusion equation

$$\frac{\partial c}{\partial t} = D\Delta c \qquad \text{(diffusion eq)}$$
$$\frac{\partial c}{\partial t} = D\Delta c + \sigma \qquad \text{(reaction-diffusion eq)}$$

where D is the diffusion coefficient.

This begs the question, what are the units of D?

$$\frac{c}{\text{time}} = D \cdot \frac{c}{\text{space}^2}$$
$$D = \frac{\text{space}^2}{\text{time}}$$

$\boxed{\text{ex}}$ How quickly can a cell get oxygen from blood? (for O_2, $D \approx 0.2\text{cm}^2/\text{sec}$)

1. The approximate distance through which diffusion works in a given time is $\propto \sqrt{Dt}$
2. The approximate time it takes to diffuse a distance d is $\propto \frac{d^2}{D}$

Time it takes O_2 to diffuse

Distance	Diffusion time
1 μm	10^{-3} sec
10 μm	0.1 sec
1 mm	10^3 sec $\approx$ 15 mins
1 cm	10^5 sec $\approx$ 25 hr
1 m	10^8 sec $\approx$ 27 yrs

Role of diffusion in cancer:

Cells in the core of the tumor die due to lack of oxygen, forming a necrotic core. There was a paper by Judah Folkman in 1971, that a tumor can only grow up to 2mm in diameter (physics). This leads to tumors developing methods to deliver oxygen to its core like angiogenesis.

Consider the diffusion equation in 1d:

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2},\ 0 < x < l,\ t > 0$$

We wish to find $u(x,t)$. Set the initial condition to $u(x,0) = f(x)$. Set the boundary condition to $u(0,t) = u(l,t) = 0,\ t > 0$, this is also known as Dirichlet boundary condition.

Alternatively, we could have the Neumann boundary condition, which states that $\frac{\partial}{\partial x}u(0,t) = \frac{\partial}{\partial x}u(l,t) = 0$. This means no flux boundary (insulating).

$\boxed{\text{ex}}$ Consider $\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2},\ 0 < x < 1,\ t > 0$. With initial condition $u(x,0) = f(x) = 2x^2$. With boundary condition $\begin{cases} u(0,t) = 0 \\ u(1,t) = 2 \end{cases}$. Find the long term behavior $\lim_{t\to\infty} u(x,t)$.

<u>Assume</u> that as $t \to \infty$, the solution comes to a fixed profile.

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} = 0$$

$$\text{As } t \to \infty,\ \frac{\partial u}{\partial t} = 0$$

We have a boundary value problem: $\begin{cases} D\frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow u(x) = cx + b \\ u(0) = 0 \\ u(1) = 2 \end{cases}$

$$\frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow \frac{dV}{dx} = 0 \Rightarrow V(x) = c$$

Use boundary condition:

$$\begin{aligned} u(0) &= c \cdot 0 + b = 0 \Rightarrow b = 0 \\ u(1) &= c \cdot 1 + 0 = 2 \Rightarrow c = 2 \\ \Rightarrow u(x) &= 2x \end{aligned}$$

To solve PDEs, we need additional techniques.

Suppose $f(x)$ defined on $[0,\pi]$. This function can be represented as a sum of sines and cosines.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where $b_n = \frac{2}{\pi}\int_0^\pi f(x)\sin nx dx$ and

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where $a_n = \frac{2}{\pi}\int_0^\pi f(x)\cos nx dx$.
This is known as sine and cosine series representation.

1.4 Lecture 4 - Solving The Diffusion Equation

Given

$$\begin{aligned} \frac{\partial u}{\partial t} &= D\frac{\partial^2 u}{\partial x^2} \qquad (1)\\ 0 < x &< \pi \\ t &> 0 \end{aligned}$$

and the following IC and BCs (Dirichlet)

$$\begin{aligned} u(x,0) &= f(x) = x(\pi - x) \qquad (2)\\ u(0,t) &= 0 \qquad (3)\\ u(\pi,t) &= 0 \qquad (4) \end{aligned}$$

<u>Use the Method of Separation of variables.</u>
Goal: find $u(x,t)$
Assumption: $u(x,t) = X(x)\cdot T(t)$
Substitute into (1)

$$\begin{aligned} \frac{\partial}{\partial t}(XT) &= D\frac{\partial^2}{\partial x^2}(XT) \\ XT' &= DTX'' \\ \frac{T'}{DT} &= \frac{X''}{X} = -\lambda \text{ const} \end{aligned}$$

Using $(3), (4)$:

$$\begin{aligned} u(0,t) &= X(0)\cdot T(t) = 0 \\ &\Rightarrow X(0) = 0 \end{aligned}$$

From (4):

$$\begin{aligned} u(\pi, t) = X(\pi)T(t) &= 0 \\ \Rightarrow X(\pi) &= 0 \end{aligned}$$

So we have the following

$$X'' + \lambda X = 0$$
$$\begin{cases} X(0) = 0 \\ X(\pi) = 0 \end{cases} \tag{$*$}$$

For time, let us use (2):

$$\begin{aligned} u(x, 0) = X(x)T(0) &= f(x) \\ \text{But } T(0) &\neq \frac{f(x)}{X(x)} \end{aligned}$$

So we have the following

$$T' = -\lambda DT \quad \text{with no init cond} \tag{$**$}$$

Now consider the case where $\lambda < 0$

$$X'' - |\lambda| X = 0$$

Take $X(x) = e^{px}$

$$\begin{aligned} p^2 - |\lambda| &= 0 \\ p &= \pm\sqrt{|\lambda|} \end{aligned}$$

We have the following general solution to $X(x)$

$$\begin{aligned} X(x) &= Ae^{\sqrt{|\lambda|}x} + Be^{-\sqrt{|\lambda|}x} \\ &\text{using BC} \\ X(0) &= Ae^0 + Be^0 = 0 \\ B &= -A \\ X(\pi) &= Ae^{\sqrt{|\lambda|}\pi} - Ae^{-\sqrt{|\lambda|}\pi} = 0 \\ A\left(e^{\sqrt{|\lambda|}\pi} - e^{-\sqrt{|\lambda|}\pi}\right) &= 0 \\ e^{\sqrt{|\lambda|}\pi} &= e^{-\sqrt{|\lambda|}\pi} \\ e^{2\sqrt{|\lambda|}\pi} &= 1 \end{aligned}$$

This is only possible if $\lambda = 0$, but $\lambda < 0$.

$$\begin{aligned} &\Rightarrow A = B = 0 \\ &\Rightarrow X(x) = 0 \\ &\text{No nontrivial solution.} \end{aligned}$$

Now consider the case where $\lambda = 0$

$$\begin{aligned} X'' &= 0 \\ X(x) &= Ax + B \\ X(0) &= B = 0 \\ X(\pi) &= A\pi = 0 \Rightarrow A = 0 \\ &\text{No nontrivial solution.} \end{aligned}$$

Now consider the case where $\lambda > 0$

$$X'' + |\lambda| X = 0$$

We have the following general solution

$$\begin{aligned} X(x) &= A\cos(\sqrt{\lambda}x) + B\sin(\sqrt{\lambda}x) \\ X(0) &= A = 0 \\ X(\pi) &= B\sin(\sqrt{x}\pi) = 0 \end{aligned}$$

Can we have $\sin(\sqrt{\lambda}\pi) = 0$?
If $\sqrt{\lambda}\pi = k\pi \Rightarrow \sin(\sqrt{\lambda}\pi) = 0$!

$$\begin{aligned} \sqrt{\lambda} &= k, \;\; k = 1, 2, 3, \dots \\ \lambda_k &= k^2 \end{aligned}$$

Realize that nontrivial solution are possible if $\lambda_k = k^2, \; k = 1, 2, 3, \dots,$ where λ_k are eigenvalues.

$$\begin{aligned} X(x) &= B\sin(\sqrt{\lambda}\pi) \\ X_k(x) &= \underbrace{B\sin(k\pi)}_{\text{eigenfunction}} \end{aligned}$$

Recall $(**)$

$$\begin{aligned} T' &= -\lambda DT \\ &= -\lambda_k DT \\ &= -k^2 DT \\ T_k(t) &= T_0 e^{-k^2 DT} \end{aligned}$$

These satisfy $(1), (3), (4)$ but not (2), therefore these are partial solutions.

$$U_k(x,t) = X(x) \cdot T(t) = \text{const} \cdot \sin(kx) \cdot e^{-Dk^2 T}$$

Look for the full solution as

$$\begin{aligned} u(x,t) &= \sum_{k=1}^{\infty} C_k \cdot u_k(x,t) \\ &= \sum_{k=1}^{\infty} C_k \cdot \sin(kx) \cdot e^{-k^2 DT} \end{aligned}$$

where C_k is unknown.

We need to find C_k such that $u(x,0) = f(x) = x(\pi - x)$

$$u(x,0) = \sum_{k=1}^{\infty} C_k \sin(kx) \cdot 1 = f(x)$$

Find C_k

$$\begin{aligned} C_k &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin(kx) dx \\ &= \frac{2}{\pi} \int_0^{\pi} x(\pi - x) \sin(kx) dx \\ &= \frac{4(1 - \cos(k\pi))}{k^3 \pi} \\ &= \begin{cases} 0 & k \text{ even} \\ \frac{8}{k^3 \pi} & k \text{ odd} \end{cases} \end{aligned}$$

Therefore,

$$u(x,t) = \sum_{k=1}^{\infty} C_k \sin(kx) e^{-k^2 DT}$$
$$\Rightarrow \lim_{T\to\infty} u(x,t) = 0$$

Try solving the diffusion equation using Neumann BC.

$$\frac{\partial u}{\partial x}(0,t) = \frac{\partial u}{\partial x}(\pi,t) = 0$$

We have

$$\begin{cases} X'' + \lambda X = 0 \\ X'(0) = 0 \\ X'(\pi) = 0 \end{cases} \tag{$*$}$$

This gives an eigenvalue problem:

1. $\lambda < 0 \Rightarrow$ no nontrivial solution.
2. $\lambda = 0$, $X'' = 0$

$$\begin{aligned} X(x) &= Ax + B \\ X'(x) &= A \\ X'(0) &= A = 0 \\ X'(\pi) &= A = 0 \end{aligned}$$

   Here we do have a nontrivial solution: $X(x) = B = \text{const}$

3. $\lambda > 0$, $X(x) = A\cos\sqrt{\lambda}x + B\sin\sqrt{\lambda}x$

$$\begin{aligned} X'(x) &= -\sqrt{\lambda}A\sin\sqrt{\lambda}x + \sqrt{\lambda}B\cos\sqrt{\lambda}x \\ X'(0) &= -\sqrt{\lambda}A\cdot 0 + \sqrt{\lambda}B\cdot 1 = 0 \\ &\Rightarrow B = 0 \\ X'(\pi) &= -\sqrt{\lambda}A\sin\sqrt{\lambda}\pi = 0 \\ \sin\sqrt{\lambda}\pi = 0 \text{ if } \sqrt{\lambda}\pi &= k\pi \\ &\Rightarrow \lambda = k^2 \text{ works !} \end{aligned}$$

We have the following eigenfunction:

$$X_k(x) = A\cos\sqrt{\lambda}x = A\cos kx$$
$$k = 0, 1, 2, \ldots$$

and the following partial solution:

$$u_k(x,t) = \cos kx \cdot e^{-k^2Dt}$$

Again this satisfies $(1), (3), (4)$. Construct

$$\begin{aligned} u(x,t) &= \sum_{k=0}^{\infty} C_k u_k(x,t) \\ &= \sum_{k=0}^{\infty} C_0 \cos(kx) e^{-k^2Dt} \end{aligned}$$

We need to have $u(x,0) = f(x)$, so we want

$$\sum_{k=0}^{\infty} C_k \cos kx = f(x)$$

Find C_k

$$\begin{aligned} C_k &= \frac{2}{\pi}\int_0^\pi f(x)\cos kx dx \\ &= \frac{2}{\pi}\int_0^\pi x(\pi - x)\cos kx dx \\ &= -\frac{2}{k^2}(\cos k\pi + 1) = \begin{cases} -\frac{4}{k^2} & k \text{ even} \\ 0 & k \text{ odd} \end{cases} \\ C_0 &= \frac{1}{2}\cdot\frac{2}{\pi}\int_0^\pi \pi(x-\pi)dx = \frac{\pi^2}{3} \end{aligned}$$

So we have

$$\begin{aligned} u(x,t) &= C_0 + \sum_{k=1}^{\infty} C_k \cos kx \cdot e^{-k^2Dt} \\ \lim_{t\to\infty} u(x,t) &= C_0 \\ &= \frac{1}{\pi}\int_0^\pi \pi(x-\pi) \\ &= \frac{1}{\pi}\int_0^\pi f(x)dx = f_{\text{avg}} \end{aligned}$$

1.5 Lecture 6 - Chemotaxis

Comparing solutions to diffusion and reaction-diffusion equations.

$$u_t = Du_{xx}$$
$$u(x,t) \propto \frac{1}{2(\pi Dt)^{\frac{1}{2}}} \exp\left(-\frac{x^2}{4Dt}\right)$$
$$\text{Spread: } x \propto \sqrt{t}$$

For the reaction-diffusion equation

$$u_t = Du_{xx} + ru$$
$$u(x,t) \propto \frac{1}{2(\pi Dt)^{\frac{1}{2}}} \exp\left(rt - \frac{x^2}{4Dt}\right)$$
$$\text{Spread: } x \propto t$$

<u>Spread of muskrats</u>. This phenomenon is in 2d. Calculate the area $A(t)$, then take $\sqrt{A(t)}$ to get a linear size. We can then plot this as a function of t, and get a positive linear relationship.

Without space, we have the following equations for growth:

$$u_t = ru \quad \text{(exponential)}$$
$$u_t = ru(1 - \frac{u}{K}) \quad \text{(logistic)}$$
$$u_t = ru(1 - \frac{u}{K}) - p\frac{u^2}{1+u^2} \quad \text{(predation)}$$

We also apply this to reaction-diffusion equation (growth with space)

$$u_t = D\Delta u + ru(1 - \frac{u}{K})$$
$$u_t = D\Delta u + ru(1 - \frac{u}{K}) - p\frac{u^2}{1+u^2}$$

<u>Chemotaxis</u>: A system of 2 reaction-diffusion equations.

- $b(x,t)$ = concentration of bacteria
- $s(x,t)$ = concentration of substrate

- $f(s)$ = growth of bacteria
- Y = yield
- k_c = bacterial death rate

We have the following system

$$b_t = \mu b_{xx} + f(s)b - k_c b$$
$$s_t = D s_{xx} - \frac{1}{Y} f(s) b$$

To solve this system analytically, we have to simplify. Consider a similar system

$$B_t = -\frac{\partial}{\partial x}(\chi B c_x) + \frac{\partial}{\partial x}(\mu B_x)$$

- $B(x,t)$ = concentration of bacteria
- $c(x,t)$ = concentration of substrate
- χ = chemotactic constant
- $J_{\text{chemo}} = \chi B c_x$ = chemotactic flux
- $J_{\text{diffusion}} = -\mu B_x$ = diffusion flux

So we have

$$B_t = -\chi(B_x c_x + B c_{xx}) + \underbrace{\mu B_{xx}}_{\text{diff}}$$

Impose boundary conditions: assume $x \in [0, L]$ and no flux at boundary ($J_{\text{chem}} + J_{\text{diff}} = 0$).

$$B_x(0,t) = B_x(L,t) = 0$$

Solve the PDE in steady state (take $B_t = 0$)

$$\frac{\partial}{\partial x}(-\chi B c_x + \mu B_x) = 0 \qquad \text{(conservation law)}$$
$$\Rightarrow -\chi B c_x + \mu B_x = \gamma = J_{\text{tot}}$$

What is γ?
We know that $J(0,t) = J(L,t) = 0 \Rightarrow J_{\text{tot}}(x,t) = 0$

$$\begin{aligned}
J_{\text{tot}} &= -\chi B c_x + \mu B_x = 0 \\
\Rightarrow \chi c_x &= \frac{\mu}{B} B_x \\
\chi c(x) &= \mu \ln B(x) + K \\
B(x) &= K \exp\left(\frac{\chi c(x)}{\mu}\right)
\end{aligned}$$

$\boxed{\text{ex}}$ 2 bacterial populations, $u(x,t)$ and $v(x,t)$. They disperse in response to the total density, $u+v$.

$$\begin{aligned}
u(x,t) \text{ has flux } q &= -k_1 \frac{\partial}{\partial x}(u+v) \\
v(x,t) \text{ has flux } w &= -k_2 \frac{\partial}{\partial x}(u+v)
\end{aligned}$$

So we have

$$\begin{aligned}
u_t &= -(qu)_x \\
v_t &= -(wv)_x
\end{aligned}$$

and

$$\begin{aligned}
u_t &= k_1 \frac{\partial}{\partial x}\left(u \frac{\partial}{\partial x}(u+v)\right) \\
v_t &= k_2 \frac{\partial}{\partial x}\left(v \frac{\partial}{\partial x}(u+v)\right)
\end{aligned}$$

Solve in steady state: $\frac{\partial}{\partial t} \to 0$

$$\begin{aligned}
u_t &= v_t = 0 \\
\frac{\partial}{\partial x}\left(u \frac{\partial}{\partial x}(u+v)\right) &= 0 \Rightarrow u \frac{\partial}{\partial x}(u+v) = c_1 \\
\frac{\partial}{\partial x}\left(v \frac{\partial}{\partial x}(u+v)\right) &= 0 \Rightarrow v \frac{\partial}{\partial x}(u+v) = c_2
\end{aligned}$$

Case 1: $c_1 = c_2 = 0$

$$u \cdot \frac{\partial}{\partial x}(u+v) = 0 \Rightarrow u = 0 \text{ or } \frac{\partial}{\partial x}(u+v) = 0$$
$$v \cdot \frac{\partial}{\partial x}(u+v) = 0 \Rightarrow v = 0 \text{ or } \frac{\partial}{\partial x}(u+v) = 0$$

Possible solutions:

1a) $u = 0$ and $v = k_2$ or $v = 0$ and $u = k_1$

2a) $u \neq 0$ and $v \neq 0 \Rightarrow \frac{\partial}{\partial x}(u+v) = 0 \Rightarrow u = K - v$

Case 2: $c_1 \neq 0, c_2 \neq 0$

$$\frac{\partial}{\partial x}(u+v) = \frac{c_1}{u} = \frac{c_2}{v}$$
$$\Rightarrow u = \frac{c_1}{c_2} v$$

We can integrate this and get

$$\left(1 + \frac{c_1}{c_2}\right) v_x = \frac{c_2}{v}$$
$$\Rightarrow \int v dv = \int \frac{c_2}{1 + \frac{c_1}{c_2}} dx$$
$$\frac{v^2}{2} = \left(\frac{c_2}{1 + \frac{c_1}{c_2}}\right) x + K$$
$$v = \left(2x \frac{c_2}{1 + \frac{c_1}{c_2}} + K\right)^{\frac{1}{2}}$$
$$\Rightarrow u = \frac{c_1}{c_2} v$$

So far we have used several ways to reduce PDEs to ODEs:

- Separation of variables / sine and cosine series (bounded domain, IVP, BVP)
- Fourier transform (infinite domain)

- Looked at steady states

$$u_t = 0 \quad \text{(convergence?)}$$

- Traveling wave solution, reduces to an ODE.

$\boxed{\text{ex}}$ Traveling wave

$$f(x) = 1 - x^2$$

Replace $x \to x - vt \equiv z$ where v is some constant. Now we have

$$f(z) = f(x - vt) \equiv F(x, t)$$

which is a function of 2 variables.

1.6 Lecture 7 - Traveling Waves

$\boxed{\text{ex}}$ Consider $f(x) = 1 - x^2$, $z = x - vt$, where v is some constant > 0.

$$F(x, t) = f(z) = 1 - (x - vt)^2$$

Take snapshots at certain time intervals:

t	$f(z)$
$t = 0$	$f(x)$
$t = 1$	$f(x - 1)$
$t = 2$	$f(x - 2)$
$\vdots$	$\vdots$

We get a traveling wave at rate v units per time t to the right. If v is negative, it moves to the left.

Consider the Reaction-diffusion equation that describes the spread of a gene in a population. Consider also 2 alleles, a and A. Call $p(x, t)$ the frequency of individuals with the a allele. We have

$$\begin{aligned} p_t &= Dp_{xx} + \alpha p(1 - p) \\ 0 \leq p(x, t) &\leq 1 \end{aligned}$$

where D is the diffusion constant, and α is the degree of advantage presented by type a. This is derived from Hardy-Weinberg genetics. This is also a 2nd order nonlinear PDE, which means it is difficult to solve. We will look for special solutions in the form of a traveling wave.

Assume that $p(x,t) = P(z)$, where $z = x - vt$ for some constant v. This reduces the PDE to an ODE.

$$\begin{aligned} f(x - vt) &= F(z) \\ f_x &= F_z z_x \\ &= F_z \\ f_{xx} &= F_{zz} \\ f_t &= F_z z_t \\ &= -vF_z \end{aligned}$$

So we have

$$-vP_z = DP_{zz} + \alpha P(1 - P)$$

Now we have a 2nd order ODE, call $\frac{dP}{dz} = -S$

$$vS = DS_z + \alpha P(1 - P)$$
$$\begin{cases} P_z = -S \\ S_z = -\frac{\alpha}{D}P(1 - P) + \frac{vS}{D} \end{cases}$$

Use nullclines:

$$\begin{aligned} &P \text{ nullclines} : S = 0 \\ &S \text{ nullclines} : -\frac{\alpha}{D}P(1 - P) + \frac{vS}{D} = 0 \\ &s = \frac{\alpha}{v}P(1 - P) \end{aligned}$$

If we plot these nullclines we get the intersection of these nullclines at

$$\begin{aligned} &(P, S) = (0, 0) \text{ and} \\ &(P, S) = (1, 0) \end{aligned}$$

We now perform stability analysis:

$$J = \begin{bmatrix} 0 & -1 \\ \frac{\alpha}{D}(1-2P) & -\frac{v}{D} \end{bmatrix}$$

For $(0,0)$, λ is real if $\frac{v^2}{D^2} - \frac{4\alpha}{D} > 0$

$$\begin{aligned} (\frac{v}{D})^2 &> \frac{4\alpha}{D} \\ \frac{v}{D} &> 2\sqrt{\frac{\alpha}{D}} \\ v &\geq 2\sqrt{\alpha D} \end{aligned} \tag{$*$}$$

Recall that the equilibrium is stable if Det$J > 0$ and Tr$J < 0$.

$$\begin{aligned} J(1,0) &= \begin{bmatrix} 0 & -1 \\ -\frac{\alpha}{D} & -\frac{v}{D} \end{bmatrix} \\ \text{Tr}J &= -\frac{v}{D} < 0 \\ \text{Det}J &= -\frac{\alpha}{D} < 0 \\ &\text{unstable !} \end{aligned}$$

There is a trajectory that connects $(1,0)$ which is unstable, with $(0,0)$, which is stable. It is called a <u>heteroclinic orbit</u>.

$$\text{Corresponds to } \begin{cases} \text{as } z \to +\infty \text{ approach (0,0)} \\ \text{as } z \to -\infty \text{ approach (1,0)} \end{cases}$$

Note here that z is NOT time.

The heteroclinic orbit is the only important orbit. If $(*)$ is violated, the eigenvalues of $(0,0)$ have an imaginary part, resulting in a stable spiral. This causes the heteroclinic orbit to cross into negative values of P, which does not make sense.

We can draw a function of $P(z)$ and z, which looks like a "negative logistic growth" function. By reintroducing $z = x - vt$, we get a moving wave, describing the spread of the advantageous allele over space and time.

With much work, one can show that the velocity of spread v, is given by

$$v = 2\sqrt{\alpha D}$$

1.6.1 Turing Instability

Consider the Belousov-Zhabotinsky reaction in 1d where there is 2 substances whose concentration is given by

$$u_1(x,t) \text{ and } u_2(x,t)$$

We have

$$\begin{cases} \dfrac{\partial u_1}{\partial t} = f_1(u_1,u_2) + D_1\dfrac{\partial^2 u_1}{\partial x^2} \\ \\ \dfrac{\partial u_2}{\partial t} = \underbrace{f_1(u_1,u_2)}_{\text{reaction}} + \underbrace{D_2\dfrac{\partial^2 u_2}{\partial x^2}}_{\text{diffusion}} \end{cases}$$

We have a system of 2 reaction-diffusion equations. Before spatial analysis, consider the non-spatial part.

$$\begin{cases} \dfrac{\partial u_1}{\partial t} = f_1(u_1,u_2) \\ \\ \dfrac{\partial u_2}{\partial t} = f_1(u_1,u_2) \end{cases}$$

$\boxed{\text{ex}}$ Consider $\begin{cases} f_1 = -u_1^3 + e^{u_2} \\ f_2 = -4u_1 + e^{u_2} \end{cases}$
Find steady states of the above system:

$$\begin{aligned} e^{u_2} &= u_1^3 \\ 4u_1 - u_1^3 &= 0 \\ u_1(4-u_1^2) &= 0 \\ u_1(2-u_1)(2+u_1) &= 0 \\ \Rightarrow u_1 &= 0, 2, -2 \end{aligned}$$

$u_1 = 0, -2$ does not make sense because we have $e^{u_2} = 4u_1$

1.7 Lecture 8 - Turing Instability

Recall our system of reaction-diffusion equations

$$\begin{cases} \dfrac{\partial u_1}{\partial t} = f_1(u_1, u_2) + D_1 \dfrac{\partial^2 u_1}{\partial x^2} \\ \dfrac{\partial u_2}{\partial t} = f_2(u_1, u_2) + D_2 \dfrac{\partial^2 u_2}{\partial x^2} \end{cases}$$

We start with a system without diffusion, find solutions, study their stability, add diffusion back, see what happens. So first we consider

$$\begin{cases} \dfrac{\partial u_1}{\partial t} = f_1(u_1, u_2) \\ \dfrac{\partial u_2}{\partial t} = f_2(u_1, u_2) \end{cases} \tag{$*$}$$

Suppose there is a solution

$$u_1(t) = u_{1b} \qquad u_2(t) = u_{2b}$$

where u_{1b} and u_{2b} are constants and are the equilibrium of $(*)$. This means that they satisfy $(*)$.

To study stability of $(*)$, we need the Jacobian evaluated at (u_{1b}, u_{2b}).
Call

$$a_{11} = \frac{\partial f_1}{\partial u_1}\Big|_{(u_{1b},u_{2b})} \qquad a_{12} = \frac{\partial f_1}{\partial u_2}\Big|_{(u_{1b},u_{2b})}$$
$$a_{21} = \frac{\partial f_2}{\partial u_1}\Big|_{(u_{1b},u_{2b})} \qquad a_{22} = \frac{\partial f_2}{\partial u_2}\Big|_{(u_{1b},u_{2b})}$$

Consider a perturbation of the equilibrium:

$$u_1(t) = u_{1b} + v_1(t) \qquad u_2(t) = u_{2b} + v_2(t)$$

Then we have $\begin{cases} \dot{v}_1 = a_{11}v_1 + a_{12}v_2 \\ \dot{v}_2 = a_{21}v_1 + a_{22}v_2 \end{cases}$

We need to calculate eigenvalues. Stability is defined by eigenvalues of matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

It is stable if $\text{Tr}A < 0$ and $\text{Det}A > 0$. Let us assume that this equilibrium is stable, that is, $\text{Tr}A < 0$ and $\text{Det}A > 0$.

How can we study the stability of the spatial system (with diffusion)?

Again, we perturb (u_{1b}, u_{2b}), but now

$$\begin{aligned} u_1(x,t) &= u_{1b} + v_1(x,t) \\ u_2(x,t) &= u_{2b} + v_2(x,t) \end{aligned}$$

Substitute this back into the original system.

$$\begin{aligned} \frac{\partial u_1}{\partial t} &= \frac{\partial}{\partial t}(u_{1b} + v_1) = \frac{\partial v_1}{\partial t} \\ \frac{\partial^2 u_1}{\partial x^2} &= \frac{\partial^2}{\partial x^2}(u_1 b + v_1) = \frac{\partial^2 v_1}{\partial x^2} \end{aligned}$$

We use Taylor series

$$\begin{aligned} f_1(u_1, u_2) &= f_1(u_{1b} + v_1, u_{2b} + v_2) \\ &\approx f_1(u_{1b}, u_{2b}) + \frac{\partial f_1}{\partial u_1} v_1 + \frac{\partial f_1}{\partial u_2} v_2 + \dots \end{aligned}$$

So we have

$$\begin{cases} \frac{\partial v_1}{\partial t} = a_{11} v_1 + a_{12} v_2 + D_1 \frac{\partial^2 v_1}{\partial x^2} \\ \frac{\partial v_2}{\partial t} = a_{21} v_1 + a_{22} v_2 + D_2 \frac{\partial^2 v_2}{\partial x^2} \end{cases}$$

Now we use the Fourier transform, with $v_1 \mapsto \hat{v}_1$ and $v_2 \mapsto \hat{v}_2$

$$\begin{aligned} \hat{v}_1(k,t) &= \int_{-\infty}^{\infty} v_1(x,t) e^{-ikx} dx \\ \hat{v}_2(k,t) &= \int_{-\infty}^{\infty} v_2(x,t) e^{-ikx} dx \end{aligned}$$

The back transform here is

$$\begin{aligned} v_1(x,t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{v}_1(k,t) e^{ikx} dk \\ v_2(x,t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{v}_2(k,t) e^{ikx} dk \end{aligned}$$

We have

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{\partial v_1}{\partial t} e^{-ikx} dx &= \frac{\partial}{\partial t} \int_{-\infty}^{\infty} v_1 e^{-ikx} dx \\
&= \frac{\partial \hat{v}}{\partial t} \\
&= \int_{-\infty}^{\infty} (a_{11} v_1 + a_{12} v_2) e^{-ikx} dx + D_1 \int_{-\infty}^{\infty} \frac{\partial^2 v_1}{\partial x^2} e^{-ikx} dx \\
&= a_{11} \int_{-\infty}^{\infty} v_1 e^{-ikx} dx + a_{12} \int_{-\infty}^{\infty} v_2 e^{-ikx} dx - D_1 k^2 \hat{v}_1 \\
&= a_{11} \hat{v}_1 + a_{12} \hat{v}_2 - D_1 k^2 \hat{v}_1
\end{aligned}$$

We now have a system of linear ODEs!

$$\begin{aligned}
\frac{\partial \hat{v}_1}{\partial t} &= a_{11} \hat{v}_1 + a_{12} \hat{v}_2 - D_1 k^2 \hat{v}_1 \\
\frac{\partial \hat{v}_2}{\partial t} &= a_{21} \hat{v}_1 + a_{22} \hat{v}_2 - D_2 k^2 \hat{v}_2
\end{aligned}$$

Our new Jacobian is

$$A(k) = \begin{bmatrix} a_{11} - D_1 k^2 & a_{12} \\ a_{21} & a_{22} - D_2 k^2 \end{bmatrix}$$

For each k, we can determine the long-term evolution (growth vs decay) of $\hat{v}(k,t)$ using the Jacobian above.

For stability, (that is, $\hat{v}(k,t) \to 0$), we need $\mathrm{Tr} A(k) < 0$ and $\mathrm{Det} A(k) > 0$. For instability, one or both should be violated.

$$\mathrm{Tr} A(k) = a_{11} - k^2 D_1 + a_{22} - k^2 D_2 < 0 \quad \underline{\text{always}}$$

The only way to make this unstable is to make $\mathrm{Det} A(k) < 0$

$$\mathrm{Det} A(k) = (a_{11} - k^2 D_1)(a_{22} - k^2 D_2) - a_{12} a_{21} \equiv F(q)$$

Call $k^2 \equiv q$.

$$\begin{aligned}
F(q) &= D_1 D_2 q^2 - (a_{11} D_2 + a_{22} D_1) q + a_{11} a_{22} - a_{12} a_{21} \\
&= D_1 D_2 q^2 - (a_{11} D_2 + a_{22} D_1) q + \mathrm{Det} A
\end{aligned}$$

To find the minimum of $F(q)$

$$F_q = 2qD_1D_2 - (a_{11}D_2 + a_{22}D_1) = 0$$
$$\Rightarrow q_{\min} = \frac{a_{11}D_2 + a_{22}D_1}{2D_1D_2}$$

We want this quantity above to be greater than 0 AND we want $F(q_{\min}) < 0$

$$F(q_{\min}) = D_1a_{22} + D_2a_{11} > 2\sqrt{D_1D_2(a_{11}a_{22} - a_{12}a_{21})}$$

This above is the condition for instability of solution (u_{1b}, u_{2b}) in the presence of diffusion. Beautiful!

A weaker but necessary condition for Turing instability.

$$D_1a_{22} + D_2a_{11} > 0$$

Recall that $a_{11} + a_{22} < 0$. With these conditions, it must follow that they have different signs.

$$D_1a_{22} - D_2|a_{11}| > 0 \Rightarrow \frac{D_2}{D_1} < \frac{a_{22}}{|a_{11}|}$$

1.8 Lecture 9 - Method of Characteristics

Consider the following PDE

$$u_t = -J_x$$

We tried $J = -D\dfrac{\partial u}{\partial x} \Rightarrow \dfrac{\partial u}{\partial t} = D\dfrac{\partial^2 u}{\partial x^2}$. Which is a second order PDE.

Now consider $J = cu$ where c is a constant, we have

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

This is the transport equation. Consider a change of variables.

$$(x,t) \to (\xi, \eta)$$
$$\xi = x + ct \qquad \eta = x - ct$$

$$\begin{aligned} u_x &= u_\xi \xi_x + u_\eta \eta_x = u_\xi \cdot 1 + u_\eta \cdot 1 \\ u_t &= u_\xi \xi_t + u_\eta \eta_t = u_\xi \cdot c - u_\eta \cdot c \end{aligned}$$

Now plug this into the PDE

$$\begin{aligned} u_t + cu_x &= 0 \\ c\,(u_\xi - u_\eta) + c(u_\xi + u_\eta) &= 0 \\ 2cu_\xi &= 0 \\ &\Rightarrow u \text{ does not depend on } \xi \\ &\Rightarrow u \text{ only depends on } \eta \end{aligned}$$

So our general solution is

$$u = f(\eta) = f(x - ct)$$

Note that this is the traveling wave problem. Now consider $f(x-ct)$, this is constant along the lines $x - ct = a$ where a is a constant. We have

$$t = \frac{x-a}{c}$$

Suppose that at $t = 0$ the solution is $u(x, 0) = e^{-|x|}$. The solution is $f(x - ct) = e^{-|x-ct|}$. How long does it take for the cusp to reach $x = x_1$? Notice that the cusp happens when $x - ct = 0$, plugging in $x = x_1$, we get

$$t = \frac{x_1}{c}$$

Consider a more general case of a transport equation

$$u_t + c(x, t)u_x = 0$$

Assume there are lines along which the solution remains constant. Express them in parametric form.

$$(x, t) \to (x(\eta), t(\eta))$$

Recall the chain rule

$$u_\eta = u_t t_\eta + u_x x_\eta$$

For the transport equation, if we take

$$\frac{\partial t}{\partial \eta} = 1 \qquad \frac{\partial x}{\partial \eta} = c$$

Then our PDE is satisfied.

Usually these problems come with an IC

$$\begin{aligned} u_t + cu_x &= 0 \\ u(x,0) &= \phi(x) \end{aligned}$$

$\boxed{\text{ex}}$ Consider $\begin{cases} u_t + xu_x = 0 \\ u(x,0) = e^{-|x|} \end{cases}$
We know that

$$u(x,t) = f(\ln|x| - t)$$

So we have

$$u(x,0) = f(\ln|x| - t) = e^{-|x|}$$

Let $a = \ln|x| - t$, so

$$\begin{aligned} e^a &= |x| \\ \exp\{-e^a\} &= e^{-|x|} \\ f(a) &= \exp\{-e^a\} \end{aligned}$$

So our specific solution is

$$u(x,t) = \exp\{-e^{\ln|x| - t}\}$$

We can also formulate an initial-boundary value problem

$$\begin{aligned} u_t + c(x,t)u_x &= 0 \\ u(x,0) &= f(x) \\ u(0,t) &= g(t) \end{aligned}$$

$\boxed{\text{ex}}$ Consider the following IBVP

$$\begin{aligned} u_t + 5u_x &= 0 \\ u(x,0) &= \cos x,\ x > 0 \\ u(0,t) &= e^{-at},\ t > 0 \end{aligned}$$

The characteristic lines are $x - 5t = a$, with $t = \frac{x-a}{5}$. Say we have to find $u(1,4)$, what char passes through (1,4)?

$$\begin{aligned} x - 5t &= \xi \\ 1 - 5 \cdot 4 &= \xi \\ \xi &= -19 \\ x - 5t &= -19 \end{aligned}$$

Does it hit the initial condition? ($t = 0$)

$$\begin{aligned} x - 5 \cdot 0 &= -19 \\ x &= -19 \end{aligned}$$

No. Does it hit the boundary condition? ($x = 0$)

$$\begin{aligned} 0 - 5t &= -19 \\ t &= \frac{19}{5} \end{aligned}$$

So we have

$$u(1,4) = e^{-a \cdot \frac{19}{5}}$$

1.9 Lecture 10 - Applications of Transport Equation

Recall the transport equation

$$\begin{cases} u_t + x^2 u_x = 0 \\ u(x,0) = \sin x \end{cases}$$

$$\frac{dt}{d\eta} = 1 \qquad \frac{dx}{d\eta} = x^2$$

So we have

$$t = \eta \qquad -\frac{1}{x} + \xi = t$$

Which characteristic line passes through the point of interest (x_0, t_0)?

$$\begin{aligned} t_0 &= -\frac{1}{x_0} + \xi \\ \xi &= t_0 + \frac{1}{x_0} \\ \Rightarrow t &= -\frac{1}{x} + t_0 + \frac{1}{x_0} \end{aligned}$$

We follow this characteristic line back to $t = 0$, and find x_*:

$$\begin{aligned} 0 &= -\frac{1}{x_*} + t_0 + \frac{1}{x_0} \\ \frac{1}{x_*} &= t_0 + \frac{1}{x_0} \\ x_* &= \frac{1}{t_0 + \frac{1}{x_0}} \end{aligned}$$

We know that at point $(x_*, 0)$, $u(x_*, 0) = \sin x_*$

$$u(x_*, 0) \sin(\frac{1}{t_0 + \frac{1}{x_0}})$$

Therefore, $u(x_0, t_0) = \sin(\frac{1}{t_0 + \frac{1}{x_0}})$

$$\Rightarrow u(x,t) = \sin(\frac{1}{t + \frac{1}{x}})$$

$\boxed{\text{ex}}$ Consider $\begin{cases} u_t + e^{-t} u_x = 5 \\ u(x,0) = \sin x \end{cases}$
We use the same method and define the characteristic lines to be solutions

$$\frac{\partial t}{\partial \eta} = 1 \qquad \frac{\partial x}{\partial \eta} = e^{-t}$$

Along the char lines, we have $\frac{\partial u}{\partial \eta} = 5 \Rightarrow u = 5\eta + f(\xi)$

$$\begin{aligned} t = \eta \qquad \frac{dx}{dt} &= e^{-t} \\ x &= -e^{-t} + \xi \\ t &= -\ln(x - \xi) \end{aligned}$$

Consider (x_0, t_0), find the value of ξ

$$\begin{aligned} x_0 &= e^{-t_0} + \xi \\ \xi &= x_0 - e^{-t_0} \end{aligned}$$

The characteristic line is

$$x = -e^{-t} + x_0 + e^{-t_0}$$

What is x_* when $t = 0$?

$$\begin{aligned} u(\eta) &= 5\eta + \xi \\ u(0) &= \sin x_* = \xi \\ u(\eta) &= 5\eta + \sin x_* \\ u(x_0, t_0) &= 5t_0 + \sin(-1 + x_0 + e^{-t_0}) \\ u(x, t) &= 5t + \sin(-1 + x + e^{-t}) \end{aligned}$$

Application: Age-structured population dynamics.
Suppose $u(a,t)$ is the population density at time t corresponding to age a.

$$\int_0^\infty u(a,t)da = u_{\text{tot}}(t) = \text{total pop at time } t.$$

This phenomenon is described by the following PDE

$$\begin{aligned} u_t + u_a &= -\gamma(a,t)u \\ u(a, 2010) &= f(a) \end{aligned}$$

where $-\gamma(a,t)u$ is attrition or the death term.

$$\frac{\partial t}{\partial \eta} = 1 \qquad \frac{\partial a}{\partial \eta} = 1$$
$$t = \eta + b$$

We want $\eta = 0$ to correspond to $t = 2010$

$$\begin{aligned} 2010 &= 0 + b \\ t &= \eta + 2010 \\ \eta &= t - 2010 \end{aligned}$$

We also get

$$\begin{aligned} a &= \eta + c \\ a &= t - 2010 + c \end{aligned}$$

Suppose we are interested in $(a_0, t_0) = (20, 2024)$, or how many 20-year-olds are there in 2024? We find the characteristic line that passes through $(20, 2024)$

$$\begin{aligned} 20 &= 2024 - 2010 + c \\ c &= 20 - 14 = 6 \\ a &= t - 2004 \end{aligned}$$

We need a_* which is the number of 6 year-olds at the 2010 census.

$$a_* = 2010 - 2004 = 6$$

Assume that $\gamma = 0$. Then

$$\begin{aligned} u(20, 2024) &= u(a_*, 2010) \\ &= u(6, 2010) \end{aligned}$$

Now we include $\gamma(a, t)$

$$\begin{cases} \frac{\partial u}{\partial \eta} = -\gamma(a_0, \eta + 2010) \\ u(0) = f(a_*) \end{cases}$$

If $\gamma =$ constant

$$\begin{aligned} u &= u_0 e^{-\gamma\eta} \\ \Rightarrow u &= f(6) e^{-\gamma\eta} \\ u &= f(6) \exp\{-\gamma(t - 2010)\} \end{aligned}$$

$\boxed{\text{ex}}$ Consider now with a birth term

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial a} = -\gamma(a,t)u$$
$$u(a,t_0) = f(a)$$
$$u(0,t) = \int_0^\infty g(a,t)u(a,t)da$$

Inviscid Burger's equation.
Recall

$$\begin{aligned} u_t &= -J_x \\ J &= cu && \text{(transport)} \\ J &= -Du_x && \text{(diffusion)} \\ J &= \frac{u^2}{2} \end{aligned}$$

So we have the following Inviscid Burger's equations or the Riemann's equation

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0$$

Suppose we have $u = u(x(t),t)$. Use direct analogy $\frac{dx}{dt} = u$.

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial t} \\ &= \frac{\partial u}{\partial x} \cdot u + \frac{\partial u}{\partial t} = 0 \end{aligned}$$

Integrate $\frac{dx}{dt} = u$. Note that u does not change along the characteristics line, therefore, we can treat u as a constant

$$\frac{dx}{dt} = u \Rightarrow x = ut + k$$

Characteristic lines are straight bu they may have different slopes.

Suppose $u(x,0) = f(x)$ which is our initial condition. The general solution is some function $G(x-ut)$.

$$G = f \Rightarrow f(x-ut) = u(x,t)$$

$\boxed{\text{ex}}$ $f(x) = ax + b \Rightarrow u(x,t) = a(x-ut) + b$.

1.10 Lecture 11 - Inviscid Burger's Equation

We have

$$u_t + u \cdot u_x = 0$$

We have the following characteristic line

$$x - ut = k$$

Suppose the initial condition is given by

$$u(x,0) = \begin{cases} 2 & x < 0 \\ 2 - x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

Suppose a char line crosses the x-axis at point $x_* = -3 \Rightarrow$ along this line we have $u(x,t) = u_0(-3) = 2$. Therefore, the slope is $\frac{1}{2}$. If $x_* \Rightarrow u_0(1) = 1 \Rightarrow$ slope is $\frac{1}{u} = 1$.

Consider the shape of $u(x,t)$ and how it changes over time. We notice that at a critical time, we stop having a function for a solution, this is called shock formation. Do we always expect shock formation in a nonlinear equation inviscid Burgers equation? We must have a region in $u_0(x)$ where u_0 decreases, where $u_0(x) = u(x,0)$.

$\boxed{\text{ex}}$ Consider

$$u_0(x) = \begin{cases} 1 & x < 0 \\ 1 + x & 0 \leq x \leq 1 \\ 2 & x \geq 1 \end{cases}$$

When the characteristics are graphed, we see no shock formation.

When can we expect to find the first shock given $u_0(x)$? A shock forms when we first observe $\frac{\partial u}{\partial x} \to \infty$. We can use the implicit solution for the transport equation.

$$u = u_0(x - ut)$$

Differentiate in x

$$\begin{aligned} u_x &= u_0'(x-ut)\cdot\frac{\partial}{\partial x}(x-ut) \\ u_x &= u_0'(x-ut)\cdot(1-u_xt) \\ u_x(1+u_0'\cdot t) &= u_0' \\ u_x &= \frac{u_0'}{1+u_0't} \end{aligned}$$

This grows to infinity if $1+u_0't=0$. From this, we can find that

$$t_* = \min_x\{-\frac{1}{u_0'(x)} \mid u_0'(x)<0\}$$

$\boxed{\text{ex}}$ Consider the following

$$\begin{aligned} \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} &= 0 \\ u(x,0)=u_0(x) &= \begin{cases} a & x\le 0 \\ b & x>0 \end{cases} \end{aligned}$$

This does not result in a shock formation but rather a rarefaction wave.

To fill the gap, define

$$u_\epsilon \begin{cases} a & x<0 \\ \frac{(b-a)x}{\epsilon}+a & 0\le x\le\epsilon \\ b & x>\epsilon \end{cases}$$

We can solve the equation by using the method of implicit functions.

$$u = u_\epsilon(x-ut)$$

We have

$$u(x,t) = \begin{cases} a & x<ta \\ \frac{x}{t} & ta<x<tb+\epsilon \\ b & x>tb+\epsilon \end{cases}$$

Now take $\epsilon \to 0$, we still get a continuous solution.

Burger's Equation.

$$\underbrace{\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x}}_{\text{Inviscid Burger's}} = \underbrace{\gamma\frac{\partial^2 u}{\partial x^2}}_{\text{viscosity}}$$

This is a 2nd order nonlinear PDE. We can not characterize all solutions, but can find a special solution via the traveling wave solution.

Consider $\xi = x - ct$, where c is an unknown constant, and look for solutions of the special form

$$u(x,t) = v(\xi) = v(x - ct)$$

We have via chain rule

$$\begin{aligned} u_t &= v'(-c) \\ u_x &= v' \cdot 1 \\ u_{xx} &= v'' \end{aligned}$$

1.11 Lecture 12 - Burger's Equation

We have

$$u_t + uu_x = \gamma u_{xx}$$

We look for solutions of the form

$$u(x,t) = v(x - ct) = v(\xi)$$

This gives us an ODE

$$-cv' - vv' = \gamma v''$$

Notice that

$$vv' = \frac{d}{d\xi}(\frac{1}{2}v^2) = \frac{1}{2} \cdot 2v \cdot \frac{dv}{d\xi}$$

So integrating wrt to ξ, we get

$$-cv - \frac{v^2}{2} + k = \gamma v'$$
$$\gamma v' = k - cv + \frac{1}{2}v^2$$

Notice that this is a 1st-order ODE and the RHS is a quadratic function of v. If the RHS has no roots, the solution will always grow. Let us assume that the RHS has 2 roots, they are

$$c \pm \sqrt{c^2 - 2k}$$
$$\text{Let} \quad a \equiv c - \sqrt{c^2 - 2k}$$
$$b \equiv c + \sqrt{c^2 - 2k}$$

$$\Rightarrow \text{RHS} \; = \frac{1}{2}(v-a)(v-b)$$
$$c = \frac{a+b}{2}, \; b > a$$

So we have

$$2\gamma v' = (v-a)(v-b)$$
$$\int \frac{2\gamma dv}{(v-a)(v-b)} = \int d\xi$$
$$\frac{2\gamma}{b-a} \ln \frac{b-v}{v-a} = \xi - \delta$$

Solving this, we get

$$v(\xi) = \frac{a \exp((b-a)(\xi-\delta)/2\gamma) + b}{\exp((b-a)(\xi-\delta)/2\gamma) + 1}$$
$$u(x,t) = \frac{a \exp((b-a)(x-ct-\delta)/2\gamma) + b}{\exp((b-a)(x-ct-\delta)/2\gamma) + 1}$$
$$= \text{ Traveling wave solution}$$

Taking the limit where $\xi \to \pm\infty$, we get

$$\lim_{\xi \to \infty} v(\xi) = a$$
$$\lim_{\xi \to -\infty} v(\xi) = b$$

What happens when our viscosity γ goes to zero?

$$\text{If } \xi > \delta \text{ and } \gamma \to 0 \Rightarrow v \to a$$
$$\text{If } \xi < \delta \text{ and } \gamma \to 0 \Rightarrow v \to b$$

2 Stochastic Processes

2.1 Lecture 12 - Intro to Stochastics

Now we move on to stochastic descriptions, where there are states of the system (discrete, like # of cells), and transitions among states that are described by probabilities.

Discrete states:

$$j = \{0, 1, \ldots, N\}$$

Discrete time means that events happen at given time points.

Transition matrix.
Define the transition matrix $P_{ij} = \mathbb{P}(i \to j)$ in one step. P_{ij} is a stochastic matrix, meaning

$$\sum_{j=0}^{N} P_{ij} = 1$$

Boundary conditions can be imposed in different ways, below is the reflecting boundary condition.

$$\begin{bmatrix} 1-p & p & & 0 \\ q & 1-p-q & p & \\ & q & 1-p-q & p \\ 0 & & q & 1-q \end{bmatrix}$$

Alternatively, we could have absorbing boundary condition.

$$\begin{bmatrix} 1 & & & 0 \\ q & 1-p-q & p & \\ & q & 1-p-q & p \\ 0 & & & 1 \end{bmatrix}$$

These are Markov chains. This means transitions only depends on finite history. The process with an absorbing boundary condition is also called the Gambler's ruin.

The <u>absorbing state</u> i:

$$\sum_{j \neq i} P_{ij} = 0 \quad \text{(cannot leave)}$$

<u>Recurrent state</u> means the system will return to the state with probability 1 sometime in the future.

We introduce φ_i^t = probability to be at state i at time t. A property that arises is

$$\sum_{i=0}^{N} \varphi_i^t = 1$$

This means our probability of being in the state space is 1. What about the probability of being in a particular state?

$$\varphi_i^{t+1} = \varphi_{i-1}^t \cdot p + \varphi_{i+1}^t \cdot q + \varphi_i^t \cdot (1 - p - q)$$

Define the vector

$$\vec{\varphi}^t = \begin{bmatrix} \varphi_0^t \\ \varphi_1^t \\ \vdots \\ \varphi_N^t \end{bmatrix}^T$$

In general, we have

$$\vec{\varphi}^{\,t+1} = \vec{\varphi}^{\,t} \cdot P$$

Suppose at time $t = 0$, we have $i = 3$. Find probabilities to be at $0, 1, \ldots, N$ at time $t = 10$. According to the problem, we have the following initial condition

$$\begin{aligned}
\vec{\varphi}^{\,0} &= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \\
\vec{\varphi}^{\,1} &= \vec{\varphi}^{\,0} P \\
\vec{\varphi}^{\,2} &= \vec{\varphi}^{\,1} P \\
&\vdots \\
\vec{\varphi}^{\,10} &= \vec{\varphi}^{\,0} P^{10}
\end{aligned}$$

In general,

$$\boxed{\vec{\varphi}^{\,t} = \vec{\varphi}^{\,0} P^t}$$

What about long-term behavior, perhaps as $t \to \infty$, the state of the system stops changing? This means we reach a steady state.

$$\begin{aligned} \vec{\varphi}^{\,t+1} &= \vec{\varphi}^{\,t} P \\ \vec{\varphi}^{\,t} &= \vec{\varphi}^{\,t} P \end{aligned}$$

Notice that here we have $\vec{\varphi}^{\,t}$ as the left eigenvector of matrix P, with the corresponding $\lambda = 1$. For stochastic matrices, 1 is the largest eigenvalue.

$\boxed{\text{ex}}$ Wright-Fisher processes. This is a model of a population that maintains a constant size. We keep track of 2 types of individuals, the wild type a and mutant A. Assume that there are $2N$ individuals. Describe the change of generations as a sampling with replacement of the current population.

<u>States</u>. We have j of 'a' and $2N - j$ of 'A'

$$j \in \{0, 1, \ldots, 2N\}$$

Construct the transition matrix:

$$\begin{aligned} \mathbb{P}(\text{pick } a) &\equiv P_j = \frac{j}{2N} \\ \mathbb{P}(\text{pick } A) &\equiv Q_j = 1 - \frac{j}{2N} \end{aligned}$$

Sample $2N$ times, we have

$$\begin{aligned} \mathbb{P}(\text{have exactly } k \text{ of '}a\text{'}) &= \binom{2N}{k} P_j^k Q_j^{2N-k} \\ &= P_{jk} \end{aligned}$$

This gives each element of our transition matrix.

$$\begin{pmatrix} \text{had } j \\ \text{of } a \end{pmatrix} \Rightarrow \begin{pmatrix} \text{got } k \\ \text{of } a \end{pmatrix}$$

If $j = 0$, $P_{0k} = 0$ if $k \neq 0$. This is an absorbing boundary.

2.2 Lecture 13 - Markov Chains

Continuing from last lecture, we have $2N+1$ total states. Therefore, our transition matrix will have $(2N+1)^2$ entries.

We now introduce mutation, where $\mathbb{P}(a \to A) = \alpha$ and $\mathbb{P}(A \to a) = \beta$. Now we have

$$P_{jk} = (1-\alpha)\frac{j}{2N} + (\beta)\frac{2N-j}{2N}$$

Now we introduce selection, where fitness of $a = 1+s$ where s is the selection coefficient and fitness of $A = 1$. Our new population of j is

$$\mathbb{P}(\text{pick } a) = \frac{(1+s)j}{(1+s)j + 2N - j}$$

Random walk. Consider a Markov chain, with state space

$$\begin{aligned} \Omega &= \{0, \ldots, N\} \\ |\Omega| &= N+1 \end{aligned}$$

At location i, we have possibility of moving left q, moving right p, staying still $1-q-p$.

Consider the case $N = 4$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ q & 1-p-q & p & 0 & 0 \\ 0 & q & 1-p-q & p & 0 \\ 0 & 0 & q & 1-p-q & p \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

With time $t \to \infty$, we are likely to end up at one of the absorbing states.

Given the initial position $i = 3$, we have the following probability vector of where we will be

$$\varphi^0 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & \ldots \end{bmatrix}$$

As $t \to \infty$, we have $\varphi^* = \varphi^* P$. This is called the stationary distribution. For a transition matrix with an absorbing boundary condition, we have

$$\varphi^* = \begin{bmatrix} \varphi_0 & 0 & \ldots & 0 & 1-\varphi_0 \end{bmatrix}$$

Proof: Here we calculate the stationary distribution for an absorbing boundary condition.

$$\begin{aligned} \varphi_0(1) + \varphi_1(q) &= \varphi_0 \\ \varphi_0(0) + \varphi_1(1-p-q) + \varphi_2(q) &= \varphi_1 \end{aligned}$$

From this we see that $\varphi_1 = \varphi_2 = 0$ and so on.

For reflecting boundary condition

$$P = \begin{bmatrix} 1-p & p & 0 & 0 & 0 \\ q & 1-p-q & p & 0 & 0 \\ 0 & q & 1-p-q & p & 0 \\ 0 & 0 & q & 1-p-q & p \\ 0 & 0 & 0 & q & 1-q \end{bmatrix}$$

We have

$$\begin{aligned} \varphi_0(1-p) + \varphi_1 q &= \varphi_0 \\ \varphi_0(p) + \varphi_1(1-p-q) + \varphi_2(q) &= \varphi_1 \end{aligned} \tag{$*$}$$

And

$$\begin{aligned} \varphi_1 q &= \varphi_0 - \varphi_0(1-p) \\ \varphi_1 q &= \varphi_0(1-1+p) \\ \varphi_1 &= \frac{p}{q}\varphi_0 \end{aligned}$$

Returning to $(*)$

$$\begin{aligned} \varphi_0 p + (1-p-q)\frac{p}{q}\varphi_0 + \varphi_2 q &= \frac{p}{q}\varphi_0 \\ \varphi_1 &= \frac{p}{q}\varphi_0 \\ \varphi_2 &= \left(\frac{p}{q}\right)^2 \varphi_0 \\ \varphi_i &= \left(\frac{p}{q}\right)^i \varphi_0 \end{aligned}$$

Consider the probability $\mathbb{P}(i \to N) = h_i = \alpha^i$. From the position $i+1 \to N$, we have $h_{i+1} = \alpha^{i+1}$ and vice versa for $i-1 \to N$. We have

$$\begin{aligned} h_i &= ph_{i+1} + qh_{i-1} + (1-p-q)h_i \\ 0 &= p\alpha^{i+1} + q\alpha^{i-1} - (p+q)\alpha^i \\ 0 &= p\alpha + \frac{q}{\alpha} - (p+q) \\ \alpha &= \frac{p}{q} \text{ or } 1 \end{aligned}$$

2.3 Lecture 14 - Neutral Drift

Consider $P_{ij} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ q & 1-p-q & p & 0 & 0 \\ 0 & q & 1-q-p & p & 0 \\ 0 & 0 & q & 1-q-p & p \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

This is an absorbing boundary condition. If we start at i, in the long term, we either get absorbed at 0 or at N.

Call h_i the probability to get absorbed at N starting at i, with $0 \leq i \leq N$. We use a single-step analysis to proceed.

$$\begin{aligned} h_i = P_{i\to i+1}h_{i+1} + P_{i\to i-1}h_{i-1} + P_{i\to i}h_i \\ \text{for } 0 < i < N \end{aligned}$$

For boundary conditions,

$$\begin{aligned} h_0 &= 0 \\ h_N &= 1 \end{aligned}$$

In terms of P_{ij} above, we have

$$\begin{aligned} h_i &= ph_{i+1} + qh_{i-1} + (1-p-q)h_i \\ 0 &= ph_{i+1} + qh_{i-1} - (p+q)h_i \end{aligned}$$

To solve, assume that $h_i = \alpha^i$, where α is an unknown constant.

$$h_{i+1} = \alpha^{i+1}, \ h_{i-1} = \alpha^{i-1}$$

So we have

$$\begin{aligned} p\alpha^{i+1} + q\alpha^{i-1} - (p+q)\alpha^i &= 0 \\ p\alpha^2 - (p+q)\alpha + q &= 0 \\ \alpha &= \frac{(p+q) \pm \sqrt{(p+q)^2 - 4pq}}{2p} \\ \alpha &= \frac{(p+q) \pm \sqrt{p^2 + 2pq + q^2 - 4pq}}{2p} \\ \alpha &= \frac{(p+q) \pm \sqrt{p^2 - 2pq + q^2}}{2p} \\ \alpha &= \frac{(p+q) \pm \sqrt{(p-q)^2}}{2p} \\ \alpha &= \frac{(p+q) \pm (p-q)}{2p} \\ \alpha &= \begin{cases} \frac{2p}{2p} = 1 \\ \frac{2q}{2p} = \frac{q}{p} \end{cases} \end{aligned}$$

So we have 2 solutions. Plug in boundary conditions to solve.

$$\begin{aligned} h_i &= A \cdot 1^i + B(\frac{q}{p})^i \\ h_0 &= A + B = 0 \Rightarrow B = -A \\ h_N &= A \cdot 1^N + B(\frac{q}{p})^N \\ &= A(1 - (\frac{q}{p})^N) = 1 \\ &\Rightarrow A = \frac{1}{1 - (\frac{q}{p})^N} \\ \Rightarrow h_i &= \frac{1 - (\frac{q}{p})^i}{1 - (\frac{q}{p})^N} \end{aligned}$$

We have two cases, if $p > q$, and $N \gg 1$

$$h_i \approx 1 - (\frac{q}{p})^i$$

If $q > p,$ and $N \gg 1$

$$(\frac{q}{p})^N \gg 1 \Rightarrow h_i = \frac{(\frac{q}{p})^i - 1}{(\frac{q}{p})^N}$$

What if $p = q$? Consider $\frac{q}{p} = 1 - \epsilon$, and take $\epsilon \to 0$.

$$\begin{aligned}\lim_{\epsilon \to 0} \frac{1-(1-\epsilon)^i}{1-(1-\epsilon)^N} &= \lim_{\epsilon \to 0} \frac{1-1+i\epsilon+\dots}{1-1+N\epsilon+\dots} \\ &= \frac{i}{N}\end{aligned}$$

Moran process. Let's apply this to biology. Consider a simple process that includes birth, death, mutations, and competition/selection called the Moran process.

We have constant population N, with two types of individuals

Type 0 $\Rightarrow$ fitness f_0 (we consider this wildtype)
Type 1 $\Rightarrow$ fitness f_1 (we consider this mutant)

Updates to this population happen as birth-death pairs.

Suppose we have j mutants, or j type 1's. We have the number of wild-types $= N - j$ and

$$j \in \{0, 1, \dots, N\}$$

Suppose an individual dies (everyone is equally likely to die). What is the probability that a mutant is chosen to reproduce?

If we have the same fitness for both types, we have

$$\mathbb{P}(\text{pick mutant}) = \frac{j}{N}$$

But if we have different fitness, we take a weighted fraction

$$\mathbb{P}(\text{pick mutant}) = \frac{f_1 j}{f_1 \cdot j + f_0(N-j)}$$

Let's construct a transition matrix. Since we have constant population, if one dies, only one reproduces, this means that $j \to j+2$ is impossible. So we can only have

$$j \to j+1,\ j \to i-1,\ j \to j$$

Calculate the probability for each scenario.

$$\begin{aligned}
\mathbb{P}(j \to j+1) &= \mathbb{P}(\text{w.t. dies}) \times \mathbb{P}(\text{mut. repr}) \\
&= \frac{N-j}{N} \times \frac{f_1 \cdot j}{f_1 \cdot j + f_0(N-j)} \\
\mathbb{P}(j \to j-1) &= \mathbb{P}(\text{mut. dies}) \times \mathbb{P}(\text{w.t. repr}) \\
&= \frac{j}{N} \times \frac{f_0(N-j)}{f_1 \cdot j + f_0(N-j)} \\
\mathbb{P}(j \to j) &= 1 - \mathbb{P}(j \to j+1) - \mathbb{P}(j \to j-1)
\end{aligned}$$

What boundary conditions are these?

$$\begin{aligned}
\text{If } j = 0,\ \mathbb{P}(0 \to 1) = 0 \Rightarrow \mathbb{P}(0 \to 0) = 1 \\
\text{If } j = N,\ \mathbb{P}(N \to N-1) = 0 \Rightarrow \mathbb{P}(N \to N) = 1
\end{aligned}$$

This is absorbing boundary conditions! We are interested in the probability of mutant takeover or mutant fixation.

Call ρ_j the probability to absorb at N starting from j. We have

$$\begin{aligned}
\rho_j &= P(j \to j+1)\rho_{i+1} + P(j \to j-1)\rho_{j-1} + (1 - P(j \to j+1) - P(j \to j-1))\rho_j \\
0 &= \frac{N-j}{N}\frac{f_1 \cdot j}{f_1 \cdot j + f_0(N-j)}\rho_{j+1} + \frac{j}{N}\frac{f_0(N-j)}{f_1 \cdot j + f_0(N-j)}\rho_{j-1} \\
&\quad - \left[\frac{N-j}{N}\frac{f_1 \cdot j}{f_1 \cdot j + f_0(N-j)} + \frac{j}{N}\frac{f_0(N-j)}{f_1 \cdot j + f_0(N-j)}\right]\rho_j \\
0 &= f_1 \cdot \rho_{j+1} + f_0\rho_{j-1} - (f_1 + f_0)\rho_j = 0
\end{aligned}$$

With the BCs $\rho_0 = 0$ and $\rho_N = 1$ Solving for ρ_i like we did with h_i, we have

$$\rho_i = \frac{1 - (\frac{f_0}{f_1})^i}{1 - (\frac{f_0}{f_1})^N}$$

We can interpret $\frac{f_0}{f_1}$ as the relative fitness.

$$\text{If } \frac{f_1}{f_0} > 1 \Rightarrow \text{advantegeous mut.}$$
$$\text{If } \frac{f_1}{f_0} < 1 \Rightarrow \text{deleterious mut.}$$
$$\text{If } \frac{f_1}{f_0} = 1 \Rightarrow \text{neutral mut. } (\rho_i = \frac{i}{N})$$

Absorption time. For absorption time, we go back to q and p.
Call $\gamma_i = E(t|\text{absorption starting at } i)$. We use 1st-step analysis.

$$\gamma_i = p\gamma_{i+1} + q\gamma_{i-1} + (1-p-q)\gamma_i + 1$$
$$0 < i < N$$

This is an inhomogeneous system. Now consider boundary conditions.

$$\gamma_i = 0$$
$$\gamma_N = 0$$

Simplify by assuming that $p = q = \frac{1}{2}$. Take

$$i = 1 : \gamma_1 = \frac{1}{2}\gamma_2 + \frac{1}{2}\gamma_0 + 1 \Rightarrow -1 = \frac{1}{2}(\gamma_2 - \gamma_1) - \frac{1}{2}(\gamma_1 - \gamma_0)$$
$$i = 2 : \gamma_2 = \frac{1}{2}\gamma_3 + \frac{1}{2}\gamma_1 + 1 \Rightarrow -1 = \frac{1}{2}(\gamma_3 - \gamma_2) - \frac{1}{2}(\gamma_2 - \gamma_1)$$
$$\vdots$$

Take $x_i = \gamma_i - \gamma_{i-1}$

$$-1 = \frac{1}{2}x_2 - \frac{1}{2}x_1$$
$$-1 = \frac{1}{2}x_3 - \frac{1}{2}x_2$$
$$\vdots$$
$$-1 = \frac{1}{2}x_N - \frac{1}{2}x_{N-1}$$

We can solve for x_i in each and plug it in the next, this is a collapsing telescopic sum, where all the sums cancel out except for x_1 and x_N.

2.4 Lecture 15 - Yule Process

Recall that the mean time to fixation is

$$\gamma_i = p\gamma_{i+1} + q\gamma_{i-1} + (1-p-q)\gamma_i + 1$$
$$\text{Take } p = q = \frac{1}{2} \text{ and } x_i = \gamma_i - \gamma_{i-1}$$

We have

$$\begin{aligned} -1 &= \frac{1}{2}x_2 - \frac{1}{2}x_1 \\ -1 &= \frac{1}{2}x_3 - \frac{1}{2}x_2 \\ &\vdots \\ -1 &= \frac{1}{2}x_N - \frac{1}{2}x_{N-1} \end{aligned}$$

We can add the first two equations

$$\begin{aligned} -2 &= \frac{1}{2}x_3 - \frac{1}{2}x_1 \\ -3 &= \frac{1}{2}x_4 - \frac{1}{2}x_1 \end{aligned}$$

In general,

$$x_k = x_1 - 2(k-1) \text{ for } k = 2, 3, \ldots, N$$

For each x

$$\begin{aligned} &x_1 = \gamma_1 - \gamma_0 = \gamma_1 \\ &x_2 = \gamma_2 - \gamma_1 \\ &x_3 = \gamma_3 - \gamma_2 = \gamma_1 - 2 \cdot 2 \Rightarrow \gamma_3 = \gamma_2 + \gamma_1 - 2 \cdot 2 \\ &x_4 = \gamma_4 - \gamma_3 = \gamma_1 - 2 \cdot 3 \Rightarrow \gamma_4 = \gamma_3 + \gamma_1 - 2 \cdot 3 \\ &\quad\Downarrow \\ &\gamma_k = k\gamma_1 - 2(1 + 2 + \cdots + (k-1)) = k\gamma_1 - k(k-1) \end{aligned}$$

We want

$$\begin{aligned} \gamma_N &= N\gamma_1 - N(N-1) = 0 \\ \gamma_1 &= N - 1 \\ \Rightarrow \gamma_k &= k(N-1) - k(k-1) = k(N-k) \end{aligned}$$

$\boxed{\text{ex}}$ Consider the demography problem in Karlin & Taylor. We have 6 stages to human life.

E_0	pre-puberty
E_1	single
E_2	married
E_3	divorced
E_4	widowed
E_5	dead

We have the following transition matrix

	E_0	E_1	E_2	E_3	E_4	E_5
E_0	0	0.9	0	0	0	0.1
E_1	0	0.5	0.4	0	0	0.1
E_2	0	0	0.6	0.2	0.1	0.1
E_3	0	0	0.4	0.5	0	0.1
E_4	0	0	0.4	0	0.5	0.1
E_5	0	0	0	0	0	1

Let ω_i be the mean time spent in E_2 starting in i. We are interested in ω_0.

Using first-step analysis

$$\begin{aligned}
\omega_0 &= 0.9\omega_1 + 0.1\omega_5 \\
\omega_1 &= 0.5\omega_1 + 0.4\omega_2 + 0.1\omega_5 \\
\omega_2 &= 1 + 0.6\omega_2 + 0.2\omega_3 + 0.1\omega_4 + 0.1\omega_5 \\
\omega_3 &= 0.4\omega_2 + 0.5\omega_3 + 0.1\omega_5 \\
\omega_4 &= 0.4\omega_2 + 0.5\omega_4 + 0.1\omega_5 \\
\omega_5 &= 0
\end{aligned}$$

Notice that this is in the form $A\vec{\omega} = B$, this matrix is not singular, therefore there is a unique solution.

Continuous-time Markov process. This is a linear birth-death process which is different from the Moran process. Birth-death events are not coupled and can happen at any time. Therefore, our states (number of cells) are discrete, but our time is continuous.

Denote $\varphi_i(t)$ = probability to have i cells at time t. There are divisions $(i \to i+1)$ and deaths $(i \to i-1)$.

If we have discrete time:

$$\begin{aligned} P(i \to i+1) &= L \cdot i \\ P(i \to i-1) &= D \cdot i \\ \text{where } L, D &\text{ are constants} \end{aligned}$$

For sufficiently large i, our probabilities make no sense. Instead, take an infinitesimally small time-interval, say Δt.

$$\begin{aligned} P(i \to i+1) &= L \cdot i \cdot \Delta t \\ P(i \to i-1) &= D \cdot i \cdot \Delta t \\ P(i \to i) &= 1 - P(i \to i+1) - P(i \to i-1) \end{aligned}$$

and

$$P(i \to k) = 0 \quad k \notin \{i, i-1, i+1\}$$

Now we perform first-step analysis

$$\begin{aligned} \varphi_i(t+\Delta t) &= \varphi_{i-1}(t) \cdot P(i-1 \to i) + \varphi_{i+1} \cdot P(i+1 \to i) + \varphi_i(t) \cdot P(i \to i) \\ &= \varphi_{i-1}(t) L(i-1)\Delta t + \varphi_{i+1}(t) D(i+1)\Delta t + \varphi_i(t)(1 - Li\Delta t - Di\Delta t) \\ \varphi_i(t+\Delta t) - \varphi_i &= \varphi_{i-1} L(i-1)\Delta t + \varphi_{i+1} D(i+1)\Delta t - \varphi_i(Li\Delta t + Di\Delta t) \\ \frac{\varphi_i(t+\Delta t) - \varphi_i(t)}{\Delta t} &= \varphi_{i-1} L(i-1) + \varphi_{i+1} D(i+1) - \varphi_i i(L+D) \end{aligned}$$

Take the limit as $\Delta t \to 0$ on both sides, we have

$$\frac{d\varphi_i}{dt} = \varphi_{i-1} L(i-1) + \varphi_{i+1} D(i+1) - \varphi_i i(L+D)$$

Consider the special case when $i = 0$?

$$\begin{aligned} \varphi_0(t+\Delta t) &= \varphi_1(t) \cdot D \cdot 1\Delta t + \varphi_0(t) \cdot 1 \\ \frac{\varphi_0(t+\Delta t) - \varphi_0(t)}{\Delta t} &= \varphi_1(t) D \\ \frac{d\varphi_0}{dt} &= \varphi_1(t) D \end{aligned}$$

So we have the following system of 1st-order linear ODE's

$$\begin{cases} \dfrac{d\varphi_i}{dt} &= \varphi_{i-1}L(i-1) + \varphi_{i+1}D(i+1) - \varphi_i i(L+D) \quad i = 1, 2, \ldots \\ \dfrac{d\varphi_0}{dt} &= \varphi_1(t)D \end{cases}$$

Consider the following initial conditions

$$\varphi_i(0) = \begin{cases} 0 & i \neq M \\ 1 & i = M \end{cases}$$

This means that at time $t = 0$, we have exactly M cells. We will solve this assuming that $D = 0$ (no death), this is the pure birth / Yule process. So we have the following problem.

$$\begin{cases} \dfrac{d\varphi_i}{dt} &= \varphi_{i-1}L(i-1) - \varphi_i Li \quad 1 < i < \infty \\ \dfrac{d\varphi_0}{dt} &= 0 \\ \varphi_i(0) &= \begin{cases} 1 & i = M \\ 0 & i \neq M \end{cases} \quad M \neq 0 \end{cases}$$

Start with $\frac{d\varphi_0}{dt} = 0 \Rightarrow \varphi_0(t) = \text{constant}$. We apply the initial condition and get $\varphi_0(0) = 0 \Rightarrow \varphi_0(t) = 0$.

So we have

$$\begin{aligned} \frac{d\varphi_1}{dt} &= L\varphi_0 \cdot 0 - L\varphi_1 \\ &= -L\varphi_1 \\ \varphi_1(t) &= ae^{-Lt} \end{aligned}$$

Given $\varphi_1(0) = 0$, we have

$$\begin{aligned} \varphi_1(0) = ae^0 &= 0 \\ a &= 0 \\ \Rightarrow \varphi_1(t) &= 0 \end{aligned}$$

Try φ_2

$$\frac{d\varphi_2}{dt} = L\varphi_1 - L\varphi_2 \Rightarrow \varphi_2(t) = 0$$

This pattern changes at $i = M$

$$\begin{aligned}\frac{d\varphi_M}{dt} &= L\varphi_{M-1}(M-1) - L\varphi_M M \\ &= -L\varphi_M M \\ &\Rightarrow \begin{cases}\varphi_M(t) = ae^{-LMt} \\ \varphi_M(0) = 1\end{cases} \\ &\Rightarrow \varphi_M(t) = e^{-LMt}\end{aligned}$$

Consider φ_{M+1} with $\varphi_{M+1}(0) = 0$

$$\begin{aligned}\frac{d\varphi_{M+1}}{dt} &= L\varphi_M \cdot M - L\varphi_{M+1}(M+1) \\ &= LMe^{-LMt} - L\varphi_{M+1}(M+1)\end{aligned}$$

We look for solutions with the form $\varphi_{M+1} = B(t)e^{-L(M+1)t}$

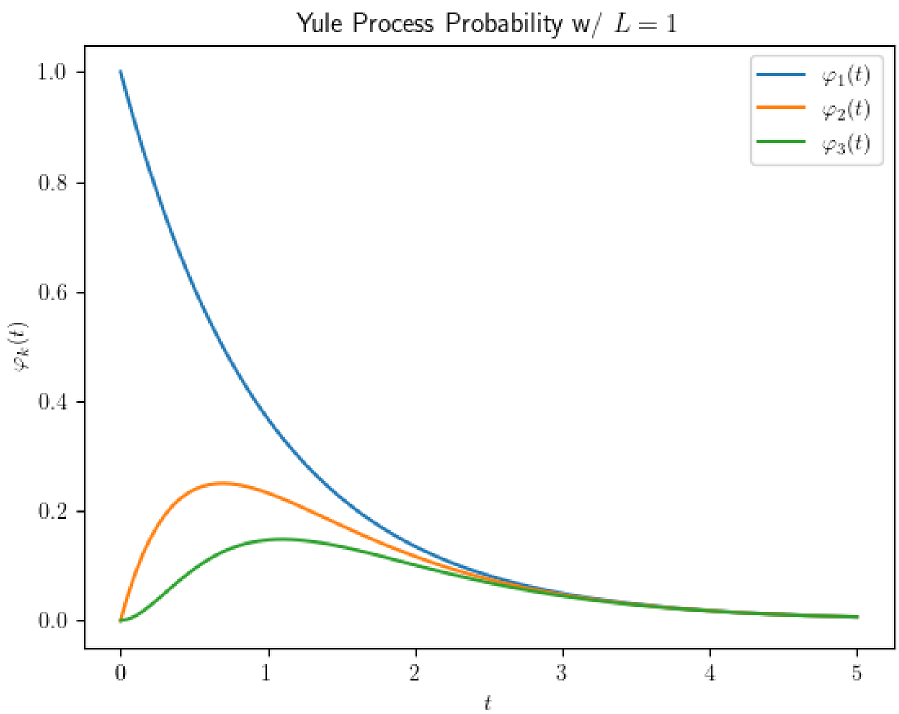
$$\begin{aligned}\frac{d\varphi_{M+1}}{dt} = \frac{dB}{dt}e^{-L(M+1)t} - L(M+1)Be^{-L(M+1)t} &= LMe^{-LMt} - L\varphi_{M+1}(M+1) \\ \frac{dB}{dt}e^{-L(M+1)t} - L(M+1)Be^{-L(M+1)t} &= LMe^{-LMt} - L(M+1)Be^{-L(M+1)t} \\ \frac{dB}{dt}e^{-L(M+1)t} &= LMe^{-LMt} \\ \frac{dB}{dt} &= LMe^{Lt(M+1-M)} \\ \frac{dB}{dt} &= LMe^{Lt} \\ B &= \int_0^t LMe^{Lt}dt \\ &= M\left[e^{Lt}\right]_0^t \\ &= M(e^{Lt} - 1)\end{aligned}$$

Therefore,

$$\begin{aligned}\varphi_{M+1}(t) &= M(e^{Lt}-1)e^{-L(M+1)t}\\ &= M(e^{Lt-Lt(M+1)} - e^{-L(M+1)t})\\ &= M(e^{-LMt} - e^{-L(M+1)t})\end{aligned}$$

Take $M = 1$, which means we had 1 cell at time $t = 0$, then

$$\boxed{\varphi_j(t) = e^{-Lt}(1-e^{-Lt})^{j-1}}$$



2.5 Lecture 16 - Linear Birth-Death Process

2.6 Lecture 17 - Linear Birth-Death Process cont.

Let $\varphi_i(t) =$ probability to have i cells at time t

$$\begin{aligned}
\frac{d\varphi_i}{dt} &= \varphi_{i-1}L(i-1) + \varphi_{i+1}D(i+1) - \varphi_i(L+D)i \qquad 1 \le i < \infty \\
\frac{d\varphi_0}{dt} &= \varphi_1 D \\
\varphi_i(0) &= \delta_{iM}
\end{aligned}$$

Can we go back from $\psi(x,t) \to \{\varphi_i(t)\}_{i=0}^{\infty}$?

Take $x = 1$

$$\begin{aligned}
\Rightarrow \psi(1,t) &= \sum_{i=0}^{\infty} \varphi_i(t) 1^i \\
&= \sum_{i=0}^{\infty} \varphi_i(t) \\
&= 1
\end{aligned}$$

Consider the first derivative of ψ

$$\begin{aligned}
\psi_x &= \frac{\partial}{\partial x} \sum_{i=0}^{\infty} \varphi_i(t) x^i \\
&= \sum_{i=0}^{\infty} \varphi_i(t) \frac{\partial}{\partial x}(x^i) \\
&= \sum_{i=1}^{\infty} \varphi_i(t) \cdot i x^{i-1}
\end{aligned}$$

Evaluate at $x = 1$

$$\begin{aligned}
\psi_x\Big|_{x=1} &= \sum_{i=1}^{\infty} \varphi_i(t) i \cdot 1^{i-1} \\
&= \sum_{i=1}^{\infty} \varphi_i(t) i \\
&= E(i)
\end{aligned}$$

Consider the second derivative of ψ

$$\begin{aligned}\psi_{xx} &= \sum_{i=0}^{\infty} \varphi_i(t) \frac{\partial^2}{\partial x^2}(x^i) \\ &= \sum_{i=2}^{\infty} \varphi_i(t) \cdot i(i-1)x^{i-2}\end{aligned}$$

Again evaluating at $x = 1$

$$\begin{aligned}\psi_{xx}\Big|_{x=1} &= \sum_{i=0}^{\infty} \varphi_i(t) i^2 - \sum_{i=0}^{\infty} \varphi_i(t) i \\ &= E(i^2) - E(i)\end{aligned}$$

And for the third derivative

$$\psi_{xxx}\Big|_{x=1} = E(i^3) - 3E(i^2) + 2E(i)$$

To get moments it is better to use moment generating function:

$$G(x,t) = \sum_{i=0}^{\infty} \varphi_i(t) e^{xi}$$

Take $x = 0$

$$\begin{aligned}\Rightarrow \psi(0,t) &= \sum_{i=0}^{\infty} \varphi_i(t) o^i \\ &= \varphi_0(t) \\ &= \text{ probability of extinction}\end{aligned}$$

Now check the first, second, and third derivatives

$$
\begin{aligned}
\psi_x\Big|_{x=0} &= \sum_{i=1}^{\infty} \varphi_i(t) i x^{i-1}\Big|_{x=0} \\
&= \sum_{i=1}^{\infty} \varphi_i(t) \cdot i \cdot 0^{i-1} \\
&= \varphi_1(t) \cdot 1 \\
\psi_{xx}\Big|_{x=0} &= \sum_{i=2}^{\infty} \varphi_i(t) i(i-1) o^{i-2} \\
&= \varphi_2(t) \cdot 2 \cdot 1 \\
\psi_{xxx}\Big|_{x=0} &= \sum_{i} \varphi_i(t) i(i-1)(i-2) 0^{i-3} \\
&= \varphi_3 \cdot 3 \cdot 2 \cdot 1
\end{aligned}
$$

We have the following pattern

$$
\begin{aligned}
\frac{\partial^n \psi}{\partial x^n}\Big|_{x=0} &= n! \varphi_n(t) \\
\varphi_i(t) &= \frac{1}{i!} \cdot \frac{\partial^i \psi}{\partial x^i}\Big|_{x=0}
\end{aligned}
$$

And so

$$
\psi(x,t) = \sum_{i=0}^{\infty} \varphi_i(t) x^i
$$